CoinKey: Multi-Factor Biometric Encryption Platform

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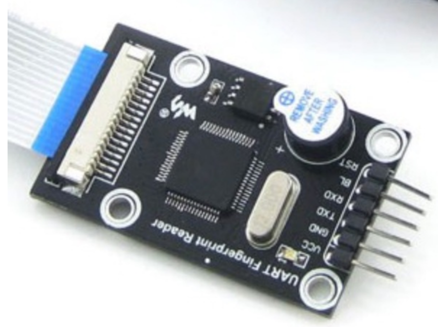
Abstract

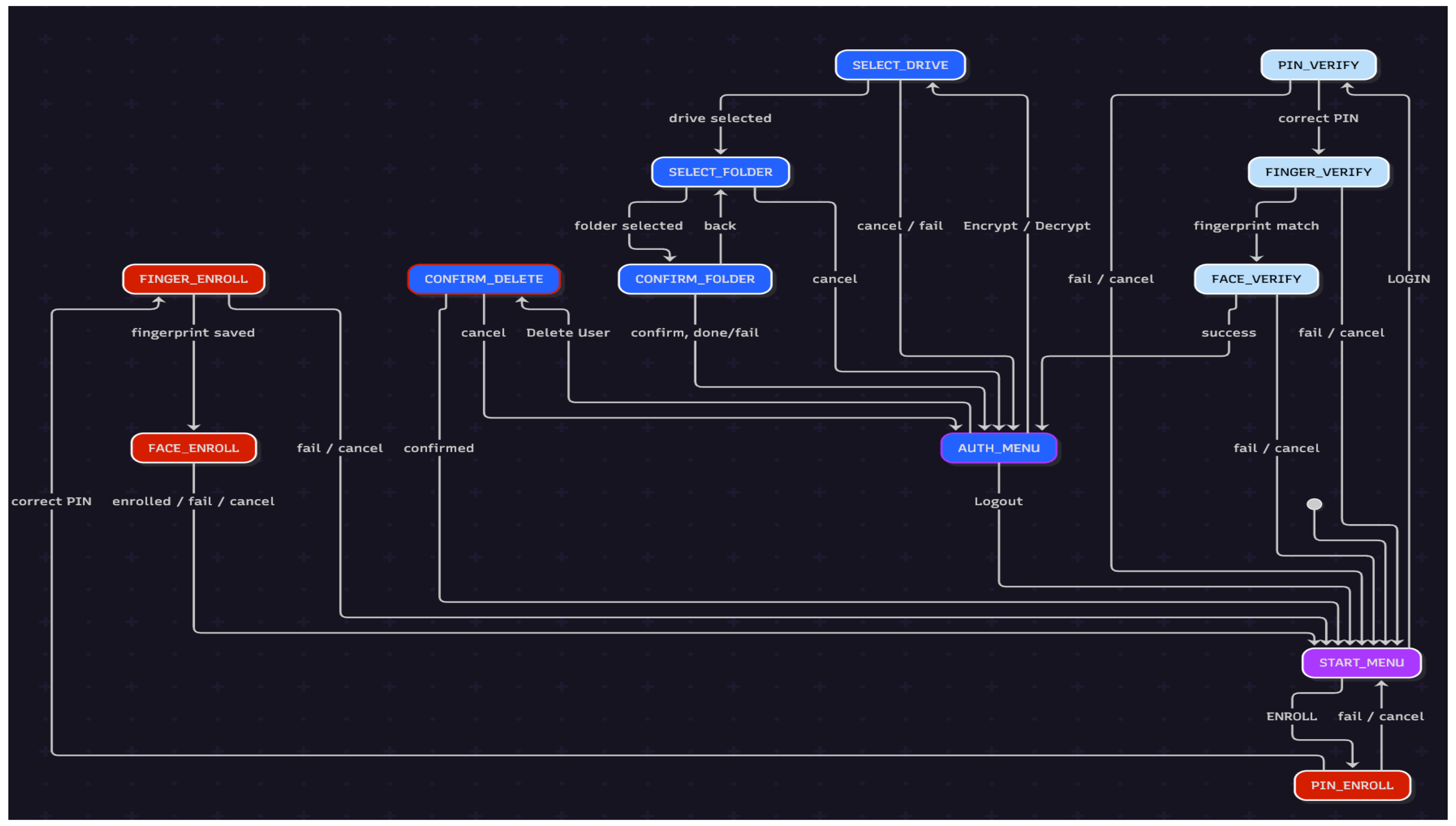
CoinKey is a standalone, offline, hardware-centric biometric authentication device designed to provide secure data encryption and storage. The system combines facial recognition, fingerprint authentication, and PIN verification to control access for encrypted files stored on general purpose USB drives. All biometric processing, credential storage, and cryptographic operations occur on device, mitigating any possible exposure to the cloud, 3rd parties, and other possible attack vectors.

Hardware

- *NVIDIA* Jetson Nano
- *ZED* 3D Depth Camera
- UART Fingerprint Reader
- 3.5in IPS LED Display
- *SparkFun* 12-key Keypad
- 5V⎓3A Power Supply
- USB-A Storage Drive



State Machine Logic



Background

- Security measures have become increasingly important as users begin to manage larger collections of sensitive data within **personal**, **professional**, and **academic** settings
- Even the largest most powerful tech organizations are at risk of regular data breaches
- Placing responsibility solely within the user's hands circumvents these known issues, lessens surface area of attack vectors, and future proofs users for the rise of **quantum** computing and the risks it poses to public key **cryptography**

Acknowledgements

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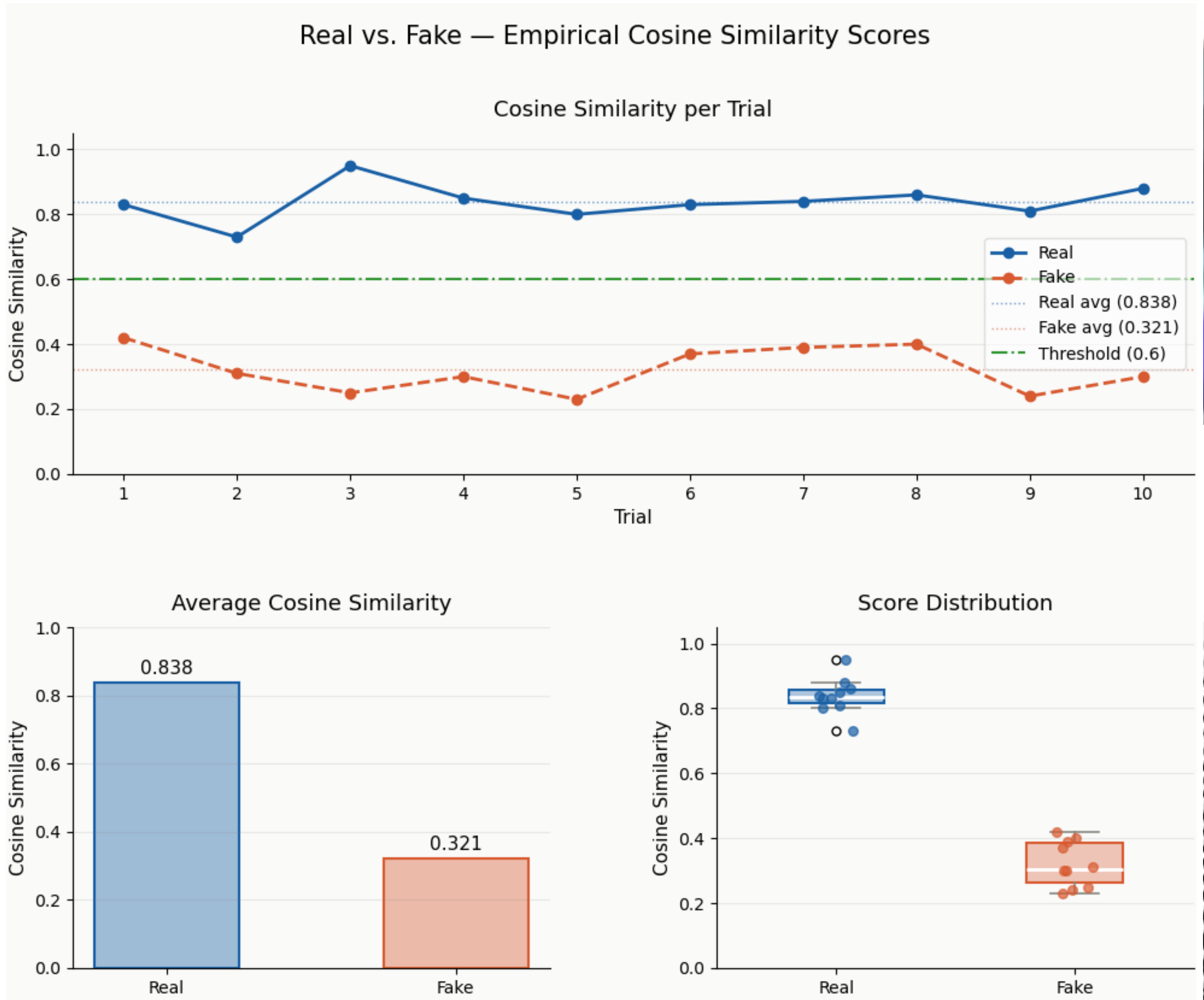
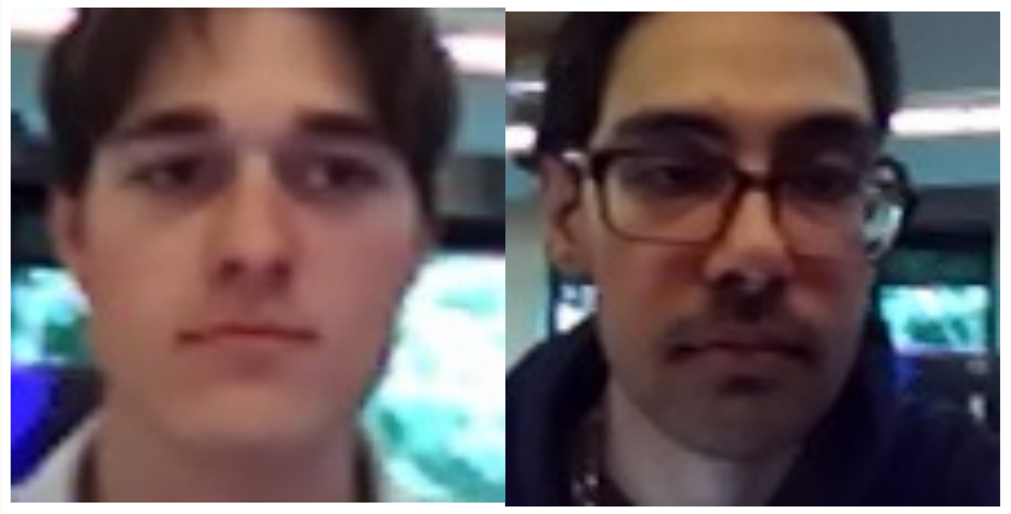
Results and Takeaways



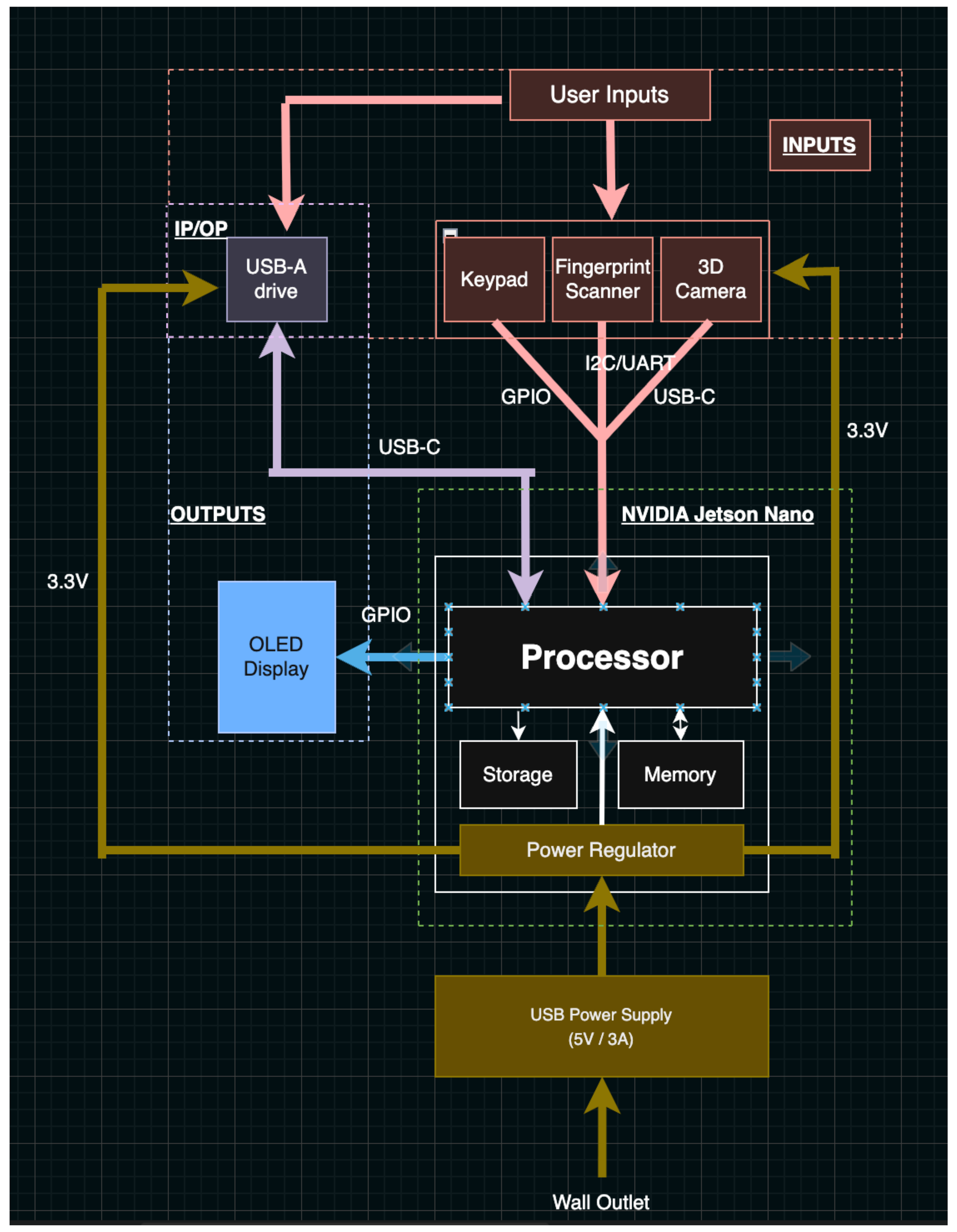
- Cosine similarity trials show an average difference of across 10 trials. As seen in the plot of Real vs Fake, the threshold is more than satisfactory for our facial scanning algorithm



- ArcFace (ResNet-50, 512-d embeddings) with similarity-transformed 112×112 face crops aligned to five facial landmarks. Cosine similarity against stored enrollment embeddings; threshold $\tau = 0.6$

- Fingerprint images of 248×296 pixels, grayness represented by 8 bits [0-255]

Hardware Design



Capstone Project

Electrical & Computer Engineering Program